Session Objectives

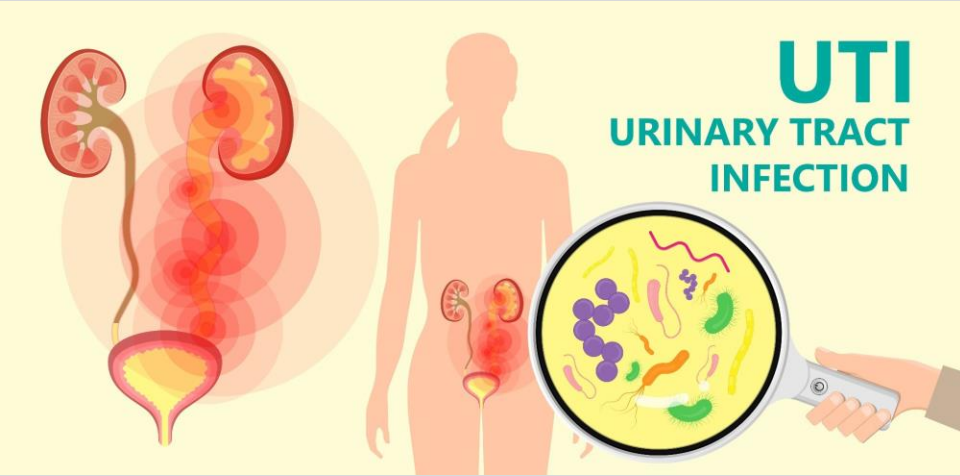
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1. Describe the epidemiology, risk factors, and terminology associated with urinary tract infections (UTIs).
2. Explain UTI pathogenesis and how it leads to different types of diseases.
3. Describe common symptoms associated with UTIs and diagnostic approaches used for identification.
4. Differentiate the pathogens that cause urinary tract infections.

Introduction

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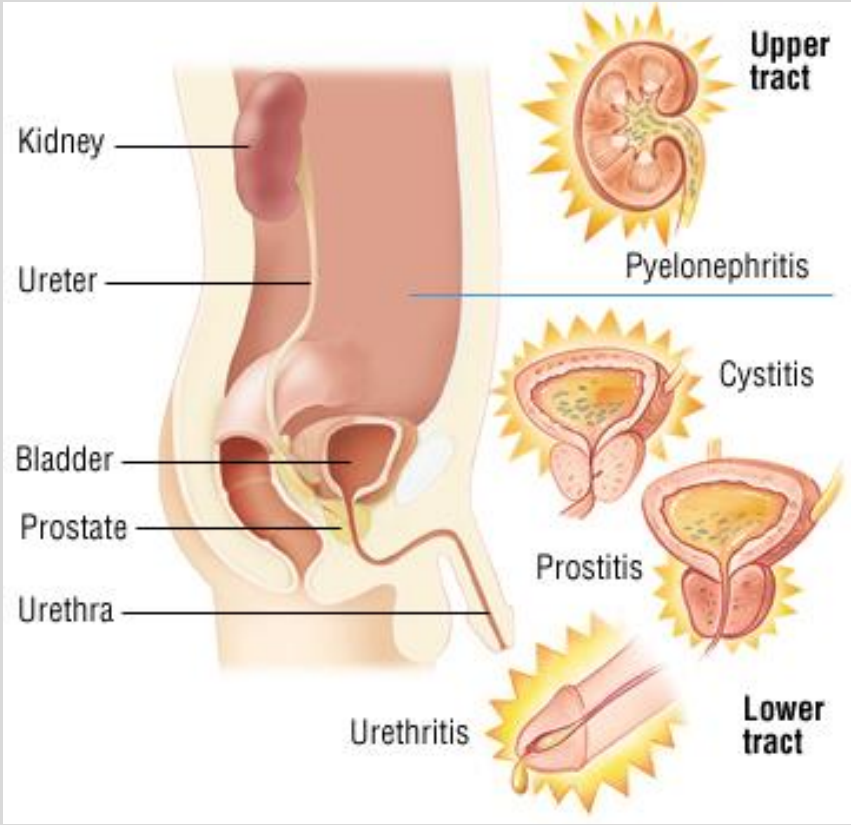
- UTIs are the most frequent outpatient infection*
- Pathogenesis
  - Colonization and ascension via the urethra
  - Subsequent ascension via the ureter
- More common in women
- Risk factors:
  - Sexual activity, age, previous UTIs, medical devices, etc.



*Medina M, Castillo-Pino E. An introduction to the epidemiology and burden of urinary tract infections. Ther Adv Urol. 2019 May 2;11:1756287219832172. doi: 10.1177/1756287219832172. PMID: 31105774; PMCID: PMC6502976.

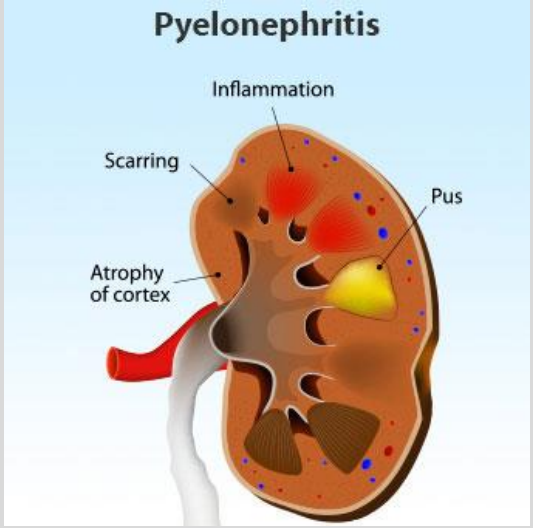
Types of UTIs

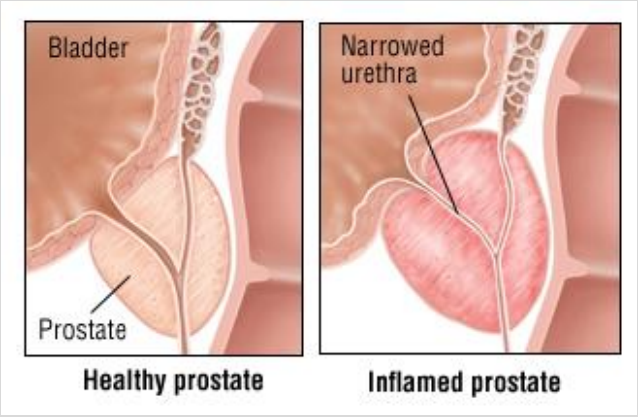
- Defined by anatomical structure:
  - Urethritis, Cystitis, Pyelonephritis, Prostatitis
- Asymptomatic vs. Symptomatic
  - *(Asymptomatic bacteriuria: asymptomatic with 2 consecutive (+) urine cultures with the same organism)*
- Complicated vs. uncomplicated
  - Uncomplicated UTI: infections in healthy women
  - Complicated UTI: all other types of UTI



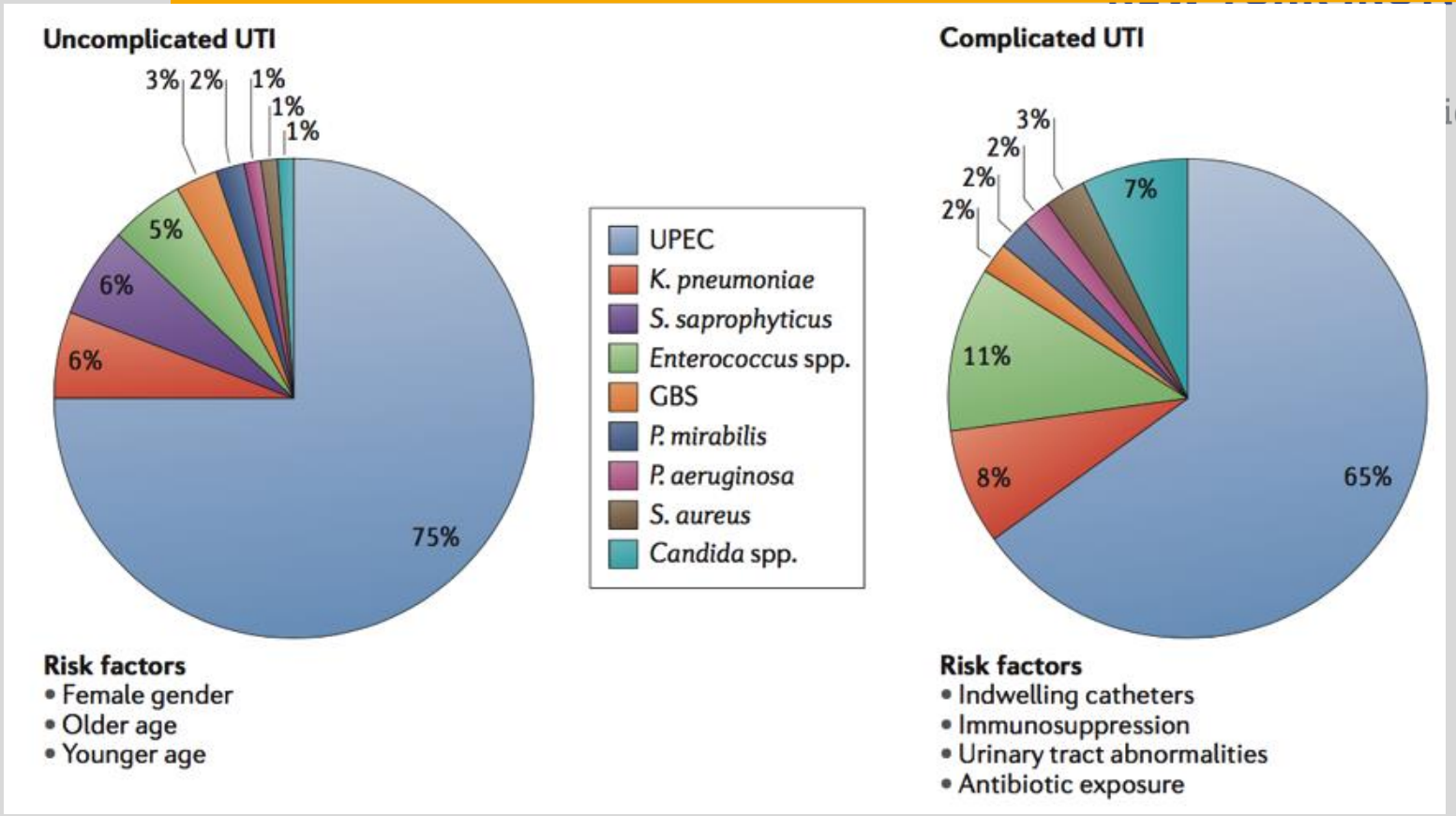
UTI Presentation

- Urethritis:
  - Gonococcal vs. Non-gonococcal (NGU)
  - Dysuria, urethral discharge, or both
- Cystitis:
  - Inflammation of the mucosal surface bladder
  - Dysuria, frequency, urgency, suprapubic pain
- Pyelonephritis:
  - Infection/inflammation of the kidney
  - Flank pain, fever, costovertebral tenderness
- Prostatitis:
  - Inflammation of the prostate
  - Acute vs chronic





Etiology of UTIs



Risk factors
- Female gender
- Older age
- Younger age

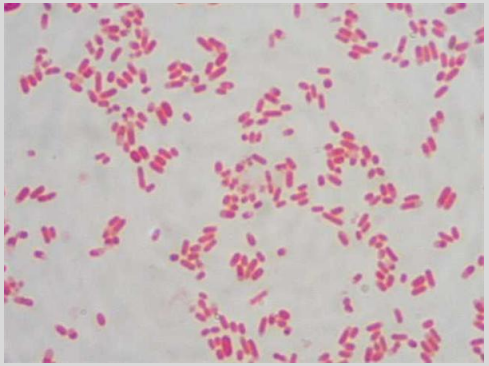
Risk factors
- Indwelling catheters
- Immunosuppression
- Urinary tract abnormalities
- Antibiotic exposure

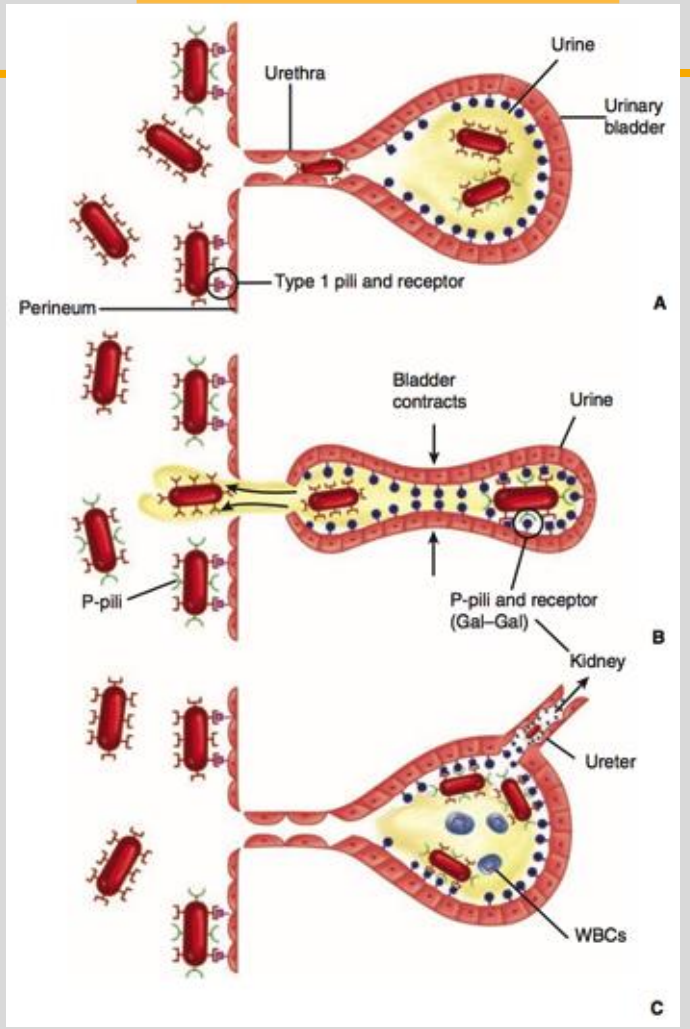
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Flores-Mireles, AL, Walker, JN, Caparon, M, Hultgren, SJ. 2015. Urinary tract infections: epidemiology, mechanisms of infection and treatment options. Nature Reviews Microbiology 13, 269–284.

Escherichia coli

- Most common cause of UTIs
- Gram negative enteric bacilli
- Distinguishing features:
  - Lactose fermenter
  - IMViC test (++--)
  - Motile
- Uropathogenic *E. coli* (UPEC)
  - Virulence factors:
    - "P"-pili and Type 1 fimbriae
    - Toxins
    - Iron acquisition
    - Capsule





Candida spp.

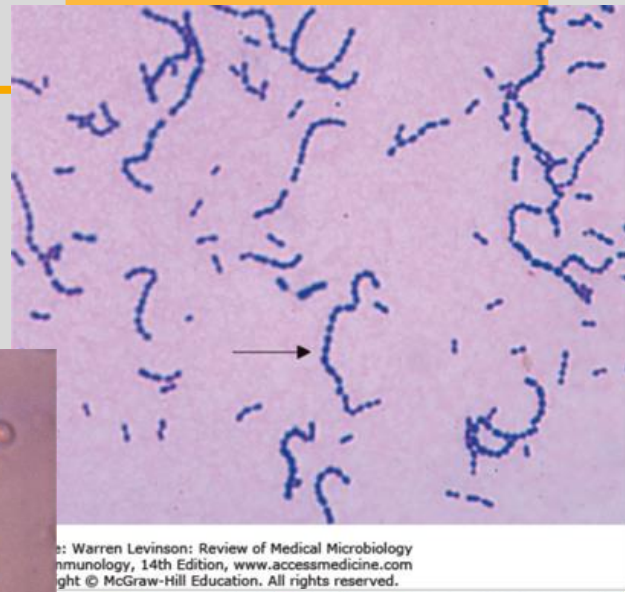
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- Opportunistic fungus
  - *C. albicans*
- Hospital associated infections
- Risk factors:
  - Medical devices
  - Antibiotic use
  - Diabetes
- Distinguishing features:
  - Pseudohyphae
  - Germ tube

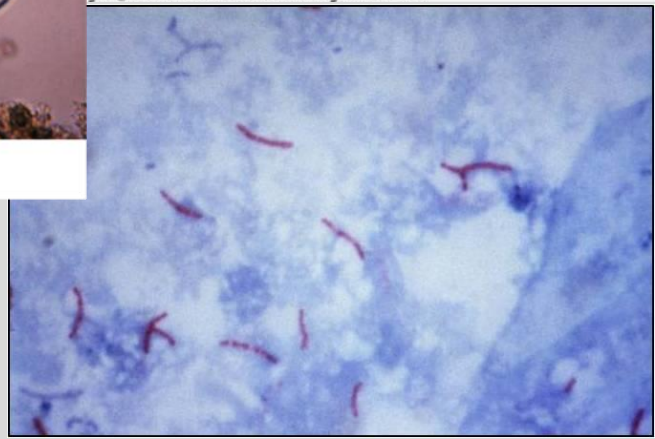


Other Possible Causes of UTIs

- *Streptococcus* spp.
  - *GBS, GAS*
- *Staphylococcus aureus*
- *Mycobacterium tuberculosis*
- Enterobacteriaceae
- *Corynebacterium urealyticum*
- *Schistosoma haematobium*
- Viruses
  - Polyomavirus, adenovirus







Summary

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- Urinary tract infections (UTIs) will present in different ways based on the type and anatomical location
- Uropathogens are usually part of the fecal flora that colonize and ascend the genitourinary tract
- The most common cause of UTIs is uropathogenic *E. coli* (UPEC)
- A small number of microbes with distinct characteristics cause the vast majority of UTIs.
- A minority of UTIs can be caused by a wide variety of microbes, including bacteria, viruses, and parasites.